
ADVERSARIAL ML: ROBUSTNESS

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1 INTRODUCTION

We are asked to train an image classifier that stays accurate under adversarial perturbation. We receive 50,000 labeled images of shape $3 \times 32 \times 32$ in 9 classes (`uint8`, no other information), and are scored on a hidden test set from the same distribution by $Score = 0.5 \cdot clean\ acc + 0.5 \cdot robust\ acc$, where robust accuracy is measured against an undisclosed attack; submissions below 50% clean accuracy are rejected. The model must be a `resnet18/34/50` producing 9 logits for $3 \times 32 \times 32$ inputs. Because neither the attack nor its parameters are revealed, the central difficulty is to build a model that is *genuinely* robust rather than one that defeats a single attack — in particular, one that does not rely on gradient masking (Athalye et al., 2018). We adopt the standard L_∞ threat model with $\epsilon = 8/255$ in $[0, 1]$ pixel space. Our full pipeline is at <https://github.com/Rhakim9374/tml-task3>.

2 APPROACH

Adversarial training. The backbone of our defense is PGD adversarial training (PGD-AT) (Madry et al., 2018): at each step we craft an L_∞ adversarial example with projected gradient descent ($\epsilon = 8/255$, step size $2/255$, 7 inner steps, random start) and minimize cross-entropy on it. PGD-AT optimizes a min-max saddle-point objective that directly targets worst-case loss; its multi-step inner maximization is a far stronger, near-worst-case first-order adversary than the single-step FGSM attack (Goodfellow et al., 2015). We train `resnet50`, the largest allowed architecture, from scratch in $[0, 1]$ pixel space. The capacity is a deliberate choice: adversarial robustness demands substantially more capacity than clean classification, since robust decision boundaries are more complex and higher-capacity networks fit the adversarial objective markedly better (Madry et al., 2018). Larger networks also have a more benign loss landscape, in which the recoverable minima concentrate in a narrow band of high-quality, well-generalizing solutions while poor minima grow exponentially rarer with width (Choromanska et al., 2015) — making the heavily regularized optimum we target easier to reach. The submitted weights are a stock `resnet50` (final layer $\rightarrow$ 9) that the server loads directly.

Regularization and stabilization. Adversarial training has a large generalization gap, so robustness is bottlenecked by regularization. Our best model combines: weight decay 10^{-3} ; an exponential moving average of the weights (EMA, decay 0.999), whose BatchNorm running statistics we recompute on clean data before evaluation (averaging weights otherwise breaks BatchNorm); light dropout (0.05) before the classifier; and gradient-norm clipping at 5 for stability. We train for 300 epochs with SGD and momentum under a cosine schedule that anneals the learning rate from a peak of 0.05 to 0 after a 5-epoch linear warm-up, which was needed to keep `resnet50` from diverging early at a high learning rate. For augmentation we use reflect-pad random crops together with random flips and 90° rotations (the dihedral D_4 group), as the classes are visually invariant to these transforms. Because the unified score weights clean and robust accuracy equally and clean accuracy was the binding side of the trade-off, we deliberately kept the inner attack light (7 steps), which preserves clean accuracy while still inducing genuine robustness.

Honest model selection. As the server’s attack is unknown, selecting checkpoints on PGD accuracy alone is dangerous: defenses that obfuscate gradients can look robust to PGD yet collapse under stronger attacks (Athalye et al., 2018). We therefore score every candidate under two attacks — a 20-step PGD and a stronger multi-restart attack combining cross-entropy with the parameter-free

DLR loss, the white-box core of AutoAttack (Croce & Hein, 2020) — and select by the *worst case* of the two. The gap between the two is our masking detector: a checkpoint whose PGD accuracy far exceeds its strong-attack accuracy is discarded even when its PGD score is high.

Techniques we evaluated but did not keep. We swept a broad set of alternatives that did not improve the final unified score; all remain reproducible behind flags in `src/train.py`. As alternative objectives we tried TRADES (Zhang et al., 2019) and MART (Wang et al., 2020): both reached higher clean accuracy but, under our strong-attack check, were partly gradient-masked and scored below PGD-AT on worst-case robustness. We tried sharpness-aware minimization (Foret et al., 2021) (no benefit at twice the cost) and AdamW in place of SGD. To curb robust overfitting we tried adversarial weight perturbation (Wu et al., 2020), a piecewise step-decay schedule (Rice et al., 2020), and an ϵ warm-up, and we tried AdvProp dual-BatchNorm (Xie et al., 2020) for the clean/adversarial split. On the regularization side we tried Cutout (DeVries & Taylor, 2017), label smoothing, stronger color jitter, and larger weight decay. None beat the plain PGD-AT + EMA + light-dropout recipe: critically, our runs never showed the robust overfitting that AWP and step-decay are designed to fix, so those levers only cost clean accuracy.

3 RESULTS

Our best submission scores **0.6273** on the public leaderboard, comfortably above the 50% clean-accuracy gate (the model keeps $\approx 72\%$ clean accuracy at $\approx 52\%$ PGD-20 robust accuracy on our held-out split). The leaderboard progression was $0.6047 \rightarrow 0.6092 \rightarrow 0.6273$: the early gains came from stabilizing `resnet50` adversarial training (EMA with BatchNorm recomputation, learning-rate warm-up), and the final jump came from a longer, gentler schedule — lowering the peak learning rate to 0.05, annealing fully to 0, and training for 300 epochs. Throughout the sweep, plain PGD-AT consistently beat TRADES and MART on worst-case robustness, because the latter’s apparent robustness was inflated by gradient masking that our strong-attack check exposed — a concrete instance of the false sense of security warned about in Athalye et al. (2018).

Notably, *none of our best runs had converged*. Their EMA validation scores increased monotonically through training and were still rising in the final epochs, with no sign of the late robust-overfitting decline that usually limits adversarial training (Rice et al., 2020) — a direct consequence of our heavy regularization and conservative attack budget. We started training late and were time-constrained at the deadline; given several more days of compute we expect the score would have continued to improve for a while. The submitted model is therefore best read as a slightly undertrained point on a still-improving curve, and its robustness/clean trade-off was tuned, not saturated.

4 CONCLUSION

Adversarial vulnerability matters because the inputs that fool a classifier are typically imperceptible to humans yet cheap for an attacker to construct, and the very gradients that train a model also tell an adversary how to break it. In safety-critical deployments — medical imaging, autonomous perception, fraud and content filtering — an attacker who can nudge pixels can induce confident, targeted errors while a human auditor sees nothing wrong, so accuracy on clean data is not a sufficient safety guarantee. Our work also underlines two practical cautions. First, there is a real clean/robust trade-off: every defense we deployed bought robustness at some cost to clean accuracy, so robustness is a deployment decision rather than a free add-on. Second, robustness must be measured against strong, diverse attacks: several of our configurations looked far more robust under a weak attack than they actually were. A model certified only against the attack it was tested on is not robust — it is merely untested against the attack that will eventually be used against it.

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